Model selection *via* marginal likelihood estimation

Robert Haobo Yuan[1,2]

[1]Department of Biosystems Science and Engineering, ETH Zürich
[2]Swiss Institute of Bioinformatics (SIB)

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- To select the best-fitting model(s) for the data.
- To test whether our results are robust to different model choices.
- To test hypotheses encoded by the models (hypothesis testing).

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- $P(\theta|\text{Data}, \text{Model})$ is the **posterior** probability of the parameter values given the data.

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- **Model**: Site and substitution model, clock model, tree model, tree topology and branching times

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- **Data**: Sequence alignments, morphological characters, fossil age information
- **Model**: Site and substitution model, clock model, tree model, tree topology and branching times
- θ: Model parameters that we put priors on and that vary during MCMC

What can we select in a model?

- Site and substitution model
- Clock model
- Tree topology and branching times
- Tree model?

- Describes how characters (*e.g.*, nucleotides and morphology) change between different states and the variation in their rates across sites/characters.

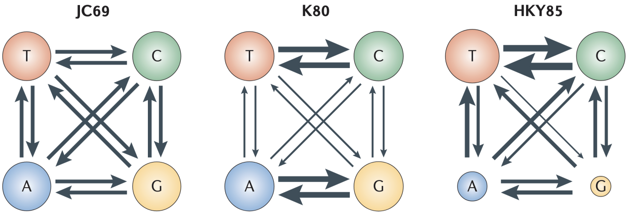

Figure: Three example substitution models with different substitution rates and equilibrium frequencies (Yang et al. 2012).

Nucleotide substitution model averaging

- Markov chain jumps between different substitution models like between parameter states in a standard MCMC.
- Parameter estimates averaged over models, weighted by their support.

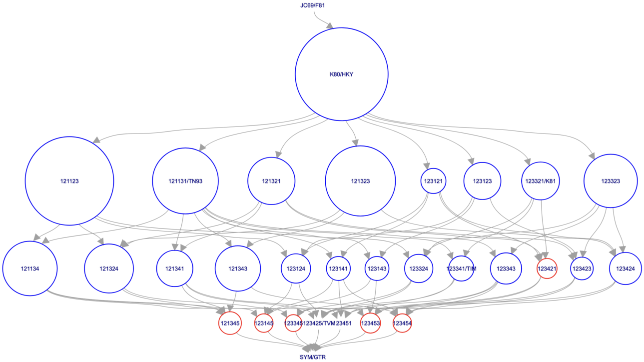

Figure: Substitution model averaging results by bModelTest (see tutorial Substitution model averaging).

Model averaging is not available for all models

Model averaging posterior:

$$P(\theta|\text{Data}) = \sum_k \overbrace{P(\text{Model}_k)}^{\text{model prior}} \underbrace{P(\text{Data}|\theta, \text{Model}_k)P(\theta|\text{Model}_k)}_{\text{posterior}}$$

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- Accounts for model uncertainty.
- **Requires specifying a model prior.**

Clock model

- Describes variation in evolutionary rates (*e.g.*, of nucleotide and morphological characters) across branches of a tree.

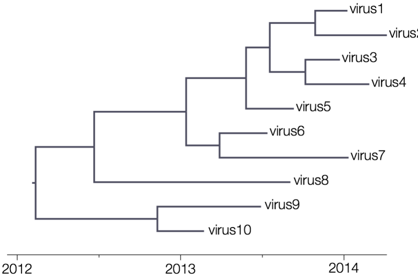


Figure: Strict clock model.

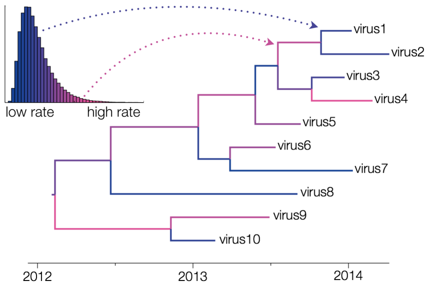
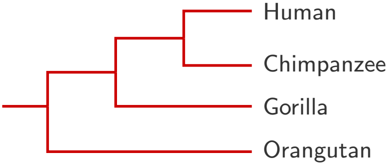


Figure: Uncorrelated LogNormal (UCLN) relaxed clock model.

Figures from https://beast.community/clocks.

Tree topology



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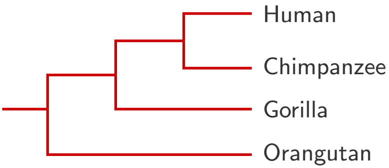

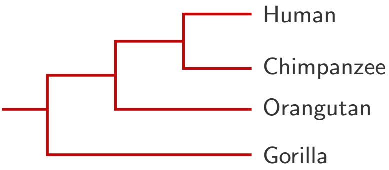

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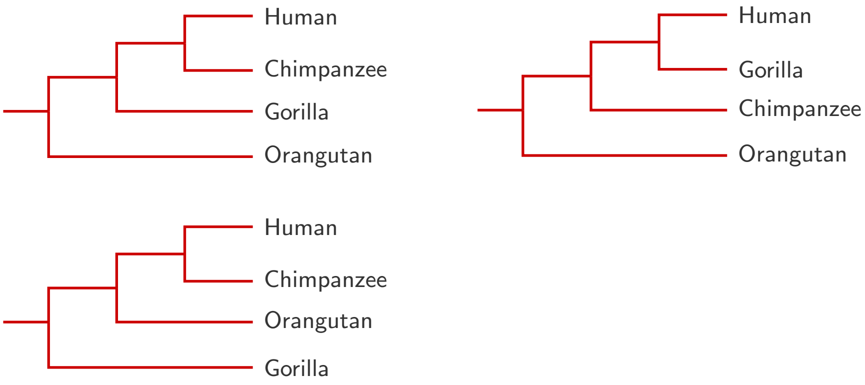

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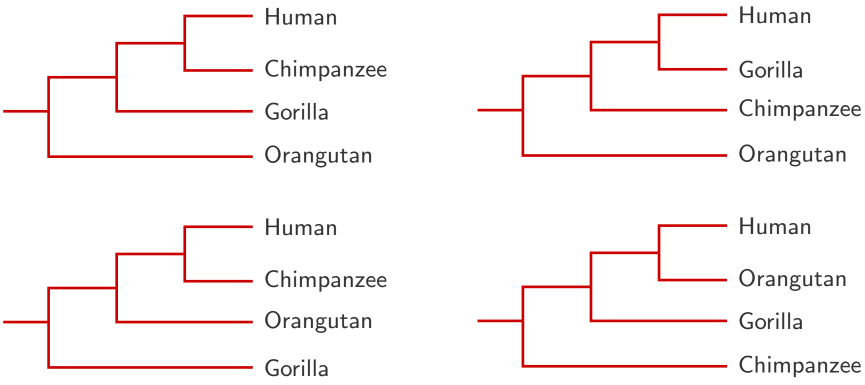

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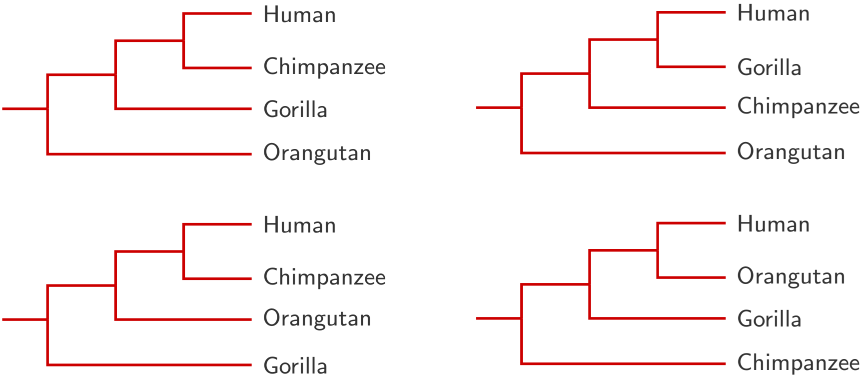
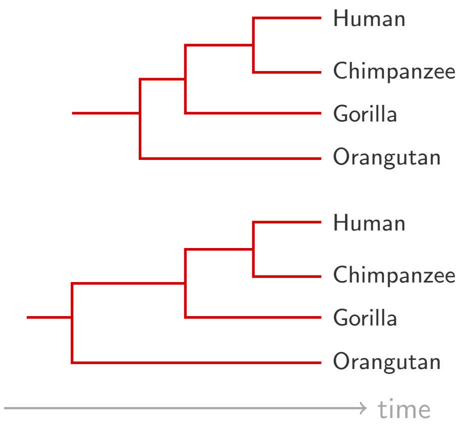


- Each tree topology is also a model.
- Model averaging of tree topologies has already been done by standard MCMC.

Branching times



- Hypothesis testing of root ages of phylogenies.
- In principle, marginal likelihood estimation could be used.

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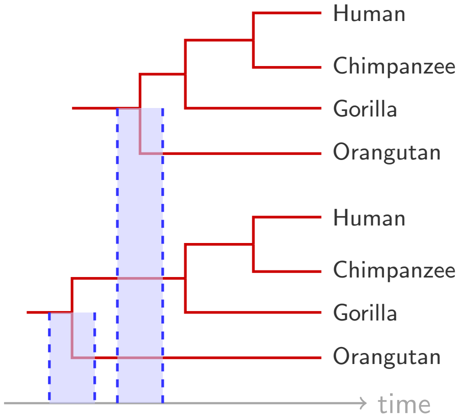

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Root age of Indo-European languages

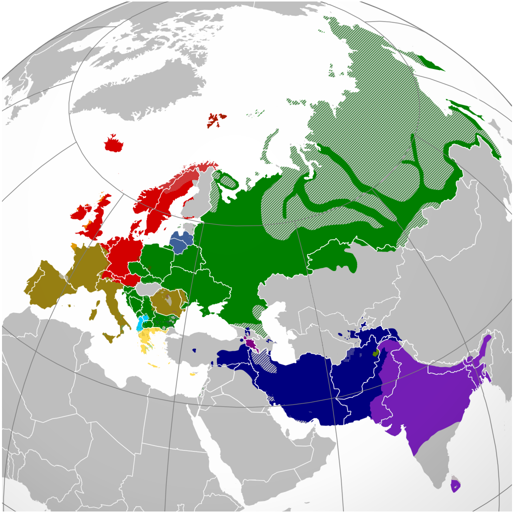

Figure: Distribution of extant Indo-European languages in Eurasia.

Root age of Indo-European languages

Example study: hypothesis testing of the root age of Indo-European (IE) languages.

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- Hypothesis A: IE languages originated in the **steppen** of modern-day Ukraine around 5-6 kya.
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- Hypothesis A: IE languages originated in the **steppen** of modern-day Ukraine around 5-6 kya.
- Hypothesis B: IE languages originated in **Anatolia** in modern-day Turkey around 8-9 kya.
- Both hypotheses can be sampled in a standard MCMC, no specific tuning required.

Root age of Indo-European languages

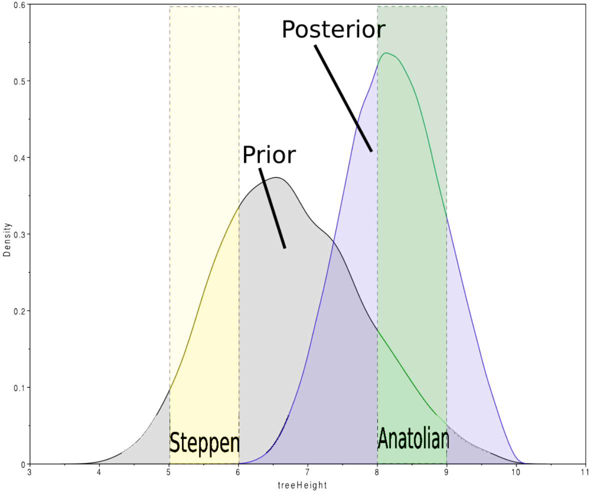

Figure: Prior and posterior distributions of root age estimates of IE languages.

Tree model?

- Describes the dynamic process that gives rise to time trees.
- Describes the distribution of tree topologies and branching times.

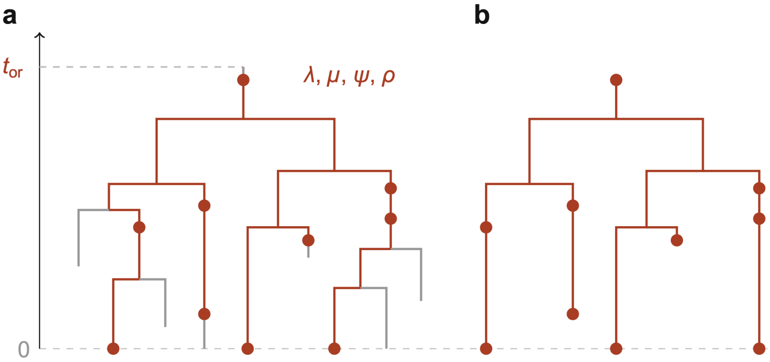

Figure: A complete and sampled tree generated by the fossilized birth–death process (Gavryushkina et al. 2020).

Tree model?

- However, standard model selection techniques (*via* marginal likelihood estimation) could not be applied (May et al. 2023).

JOURNAL ARTICLE

Diversification Models Conflate Likelihood and Prior, and Cannot be Compared Using Conventional Model-Comparison Tools FREE

Michael R May ✉ , Carl J Rothfels

Systematic Biology, Volume 72, Issue 3, May 2023, Pages 713–722, https://doi.org/10.1093/sysbio/syad010

Published: 10 March 2023 **Article history** ▾

- Use model adequacy tests instead (See "Phylodynamic Model Adequacy Using Posterior Predictive Simulations").

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Likelihood and Marginal likelihood

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Estimating ML is hard! Why?

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- We use MCMC to evade ML estimation and sample a target distribution to approximate the posterior distribution.

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    - No!
    - Parameters are defined differently across models.
    - Posteriors are only normalized within their own models, cannot be compared between models.

Naive methods for ML estimation

Arithmetic mean estimator (AME) by sampling K samples from the prior:

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- Overestimates ML, high likelihood areas overrepresented.

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Path sampling: In theory

Introduced by Lartillot et al. (2006). For a model M, we define a continuous and differentiable path joining the prior and the (unnormalized) posterior, where $0 \leq \beta \leq 1$:

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The (unnormalized) posterior:

$$q_1(\theta) = p(D|\theta, M)p(\theta|M)$$

$$Z_1 = p(D|M) = \text{Marginal likelihood}$$

Path sampling: In theory

$$\ln p(D|M) - 0 = \ln Z_1 - \ln Z_0 = \int_0^1 E_\beta[\ln p(D|\theta, M)]d\beta$$

See Lartillot et al. (2006) for details.

Path sampling: In theory

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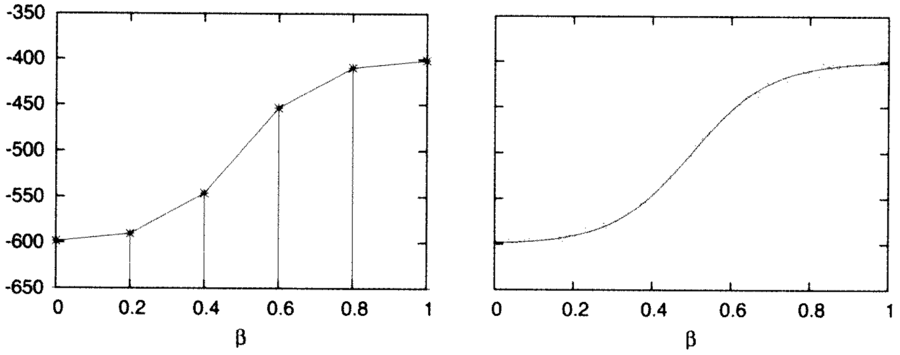

Figure: Rationale of path sampling (Lartillot et al. 2006).

Stepping-stone sampling: In theory

Introduced by Xie et al. (2011). The goal is to estimate the ratio Z_1/Z_0, which can be expressed as a product of K ratios:

$$r_{SS} = \frac{Z_1}{Z_0} = \prod_{k=1}^{K} \frac{Z_{\beta_k}}{Z_{\beta_{k-1}}},$$

where

$$Z_1 = p(D|M) = \text{Marginal likelihood}$$

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Stepping-stone sampling: In theory

Stepping-stone sampling uses sampling intervals (β values) spaced by a Beta($\alpha, 1$) distribution, where $\alpha = 0.3$ by default.

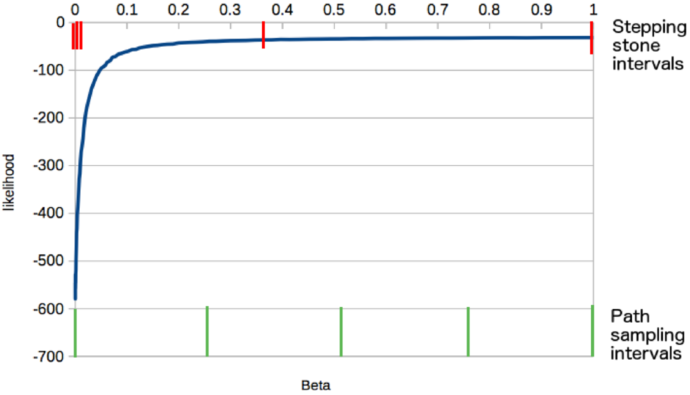

Figure: Remco R. Bouckaert's Taming the BEAST 2023 lecture.

Path and stepping-stone sampling: In practice

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- Uncertainty of estimates
  - Run multiple replicates of the same PS/SS analysis.

Path and stepping-stone sampling: In practice

Key points to consider in practice:

- MODEL_SELECTION package in BEAST2
  - Steps are resumable and parallelizable.
  - See "Path sampling" and "Hypothesis testing/model comparison".

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$$Z = \int L(\theta)\pi(\theta)d\theta = \int_0^1 L(\xi)d\xi$$

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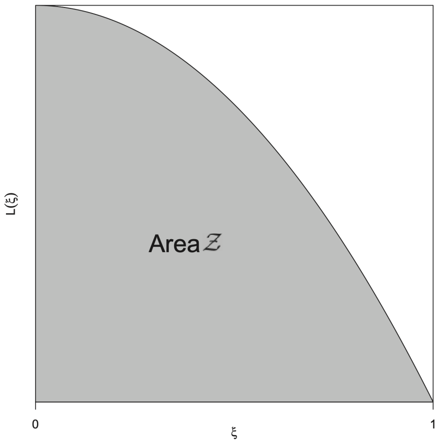

Figure: Relationship between likelihood values and the cumulative prior mass (Russel et al. 2019).

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Nested sampling: In theory

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$$Z = \int_0^1 L(\xi) d\xi$$

- ξ = proportion of prior mass with likelihood at least $L(\xi)$
- $L(\xi)$ is a monotonically decreasing function, reaching its highest point at $\xi = 0$ and its lowest at $\xi = 1$.

Nested sampling: In theory

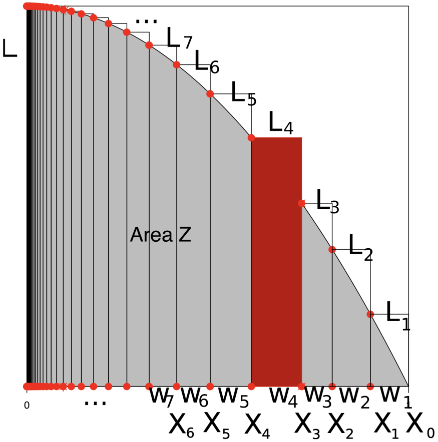

Figure: Remco R. Bouckaert's Taming the BEAST 2023 lecture.

- If a set of points on the $L(\xi)$ curve can be obtained, the integral can be approximated numerically:

$$Z \approx \sum_{i=1}^{k} w_i L_i,$$

- where $w_i = \xi_{i-1} - \xi_i$ and $L_i = L(\xi_i)$.

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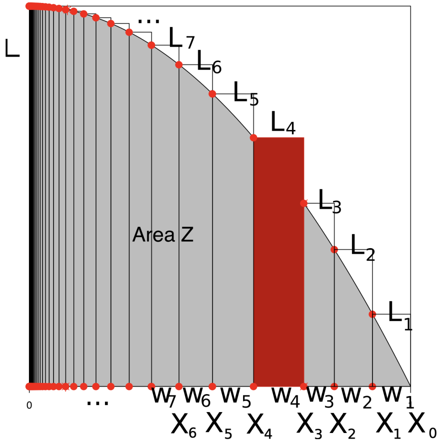

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- For a decreasing sequence of ξ values and an increasing sequence of L values, the ML can be estimated.

Nested sampling: In theory

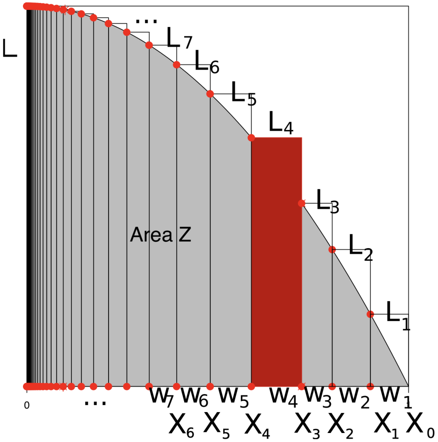

Figure: Remco R. Bouckaert's Taming the BEAST 2023 lecture.

Nested sampling (NS) works as follows:

Nested sampling: In theory

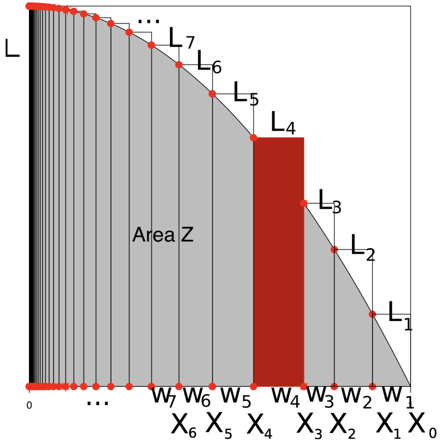

Figure: Remco R. Bouckaert's Taming the BEAST 2023 lecture.

Nested sampling (NS) works as follows:

- Randomly sample N points (particles) from the prior distribution.

Nested sampling: In theory

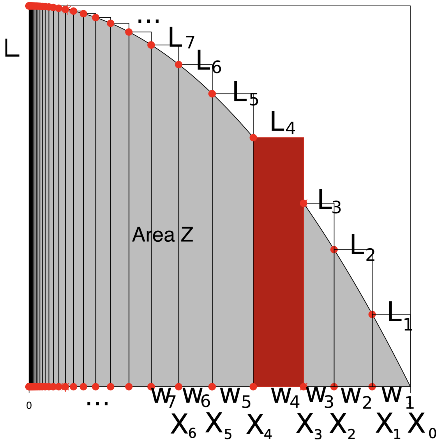

Figure: Remco R. Bouckaert's Taming the BEAST 2023 lecture.

Nested sampling (NS) works as follows:

- Randomly sample N points (particles) from the prior distribution.
- While (marginal) likelihood estimates have not converged:

Nested sampling: In theory

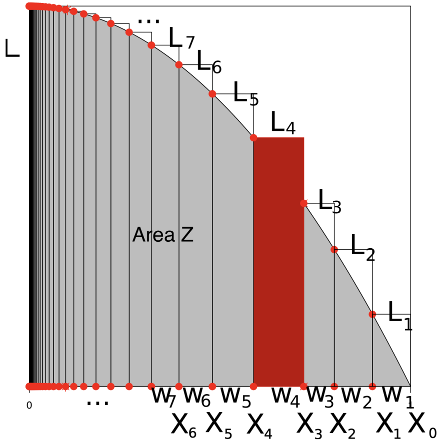

Figure: Remco R. Bouckaert's Taming the BEAST 2023 lecture.

Nested sampling (NS) works as follows:

- Randomly sample N points (particles) from the prior distribution.
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  - Pick the point with the lowest likelihood L_{min} and save it.

Nested sampling: In theory

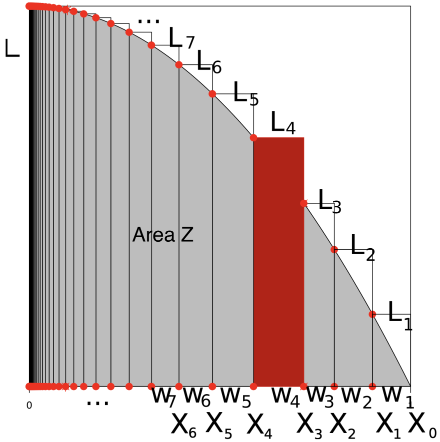

Figure: Remco R. Bouckaert's Taming the BEAST 2023 lecture.

Nested sampling (NS) works as follows:

- Randomly sample N points (particles) from the prior distribution.
- While (marginal) likelihood estimates have not converged:
  - Pick the point with the lowest likelihood L_{min} and save it.
  - Replace the point with a new one sampled from the prior via an MCMC chain of certain steps (sub-chain length), **under the condition that the new likelihood is at least L_{min}.**

Nested sampling: In practice

Key points to consider in practice:

Nested sampling: In practice

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- Number of active points

Nested sampling: In practice

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  - More active points mean more accurate estimates and less numerical uncertainty.
  - Begin your analysis with one point first, calculate the required number based on desired SDs and information.

Nested sampling: In practice

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Nested sampling: In practice

Key points to consider in practice:

- Number of active points
  - More active points mean more accurate estimates and less numerical uncertainty.
  - Begin your analysis with one point first, calculate the required number based on desired SDs and information.
- Sub-chain length
  - Short sub-chain lengths lead to inconsistent ML estimates.
  - Run multiple replicates to check whether ML estimates have converged; if not, run with longer sub-chain lengths.
  - Requires trial-and-error.

Model selection summary

- We can select for site & substitution models, clock models, tree topologies and branching times using MLE.
- Do not compare likelihoods or posteriors.
- Do not use AME or HME of ML.
- Path and stepping-stone sampling
  - Uses a path joining the prior and the unnormalized posterior.
  - Number of steps and chain length
  - **Stable and recommended**
- Nested sampling
  - Uses a relationship joining the cumulative prior mass and likelihood values.
  - Number of active points and sub-chain length

Model selection

Model selection with nested sampling
by Remco Bouckaert

Tutorial
- Github repository
- License
- Statistics

Data
- HBV.nex

XML
- HBVStrict-NS.xml
- HBVStrict-NS30.xml
- HBVStrict.xml
- HBVUCLN-NS.xml
- HBVUCLN-NS30.xml
- HBVUCLN.xml

Output
- HBVStrict-NS30.log
- HBVStrict-NS30.out
- HBVStrict.log
- HBVUCLN-NS30.log
- HBVUCLN-NS30.out
- HBVUCLN.log

Background

BEAST provides a bewildering number of models. Bayesians have two techniques to deal with this problem: model averaging and model selection. With **model averaging**, the MCMC algorithm jumps between different models, and models more appropriate for the data will be sampled more often than unsuitable models (see for example the substitution model averaging tutorial). **Model selection** on the other hand just picks one model and uses only that model during the MCMC run. This tutorial gives some guidelines on how to select the model that is most appropriate for your analysis.

Bayesian model selection is based on estimating the marginal likelihood: the term forming the denominator in Bayes' formula. This is generally a computationally intensive task and there are several ways to estimate them. Here, we concentrate on nested sampling as a way to estimate the marginal likelihood as well as the uncertainty in that estimate.

Say, we have two models, M_1 and M_2, and estimates of their (log) marginal likelihoods, Z_1 and Z_2. We can calculate the Bayes' factor (BF), which is the fraction $BF = Z_1/Z_2$ (or the difference $\log(BF) = \log(Z_1) - \log(Z_2)$ in log space). Typically, if $\log(BF)$ is greater than 1, M_1 is favoured, otherwise M_1 and $M2$ are hardly distinguishable or M_2 is favoured. How much M_1 is favoured over M_2 can be found in the following table (Kass & Raftery, 1995):

BF range	$\ln(BF)$ range	$\log_{10}(BF)$ range	Interpretation

References

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Thank you!

Thank you for your attention!